

SHETH L.U.J AND SIR MV COLLEGE

PRACTICAL NO:-08

AIM : Confidence Interval Estimation:

- Calculate confidence intervals for population means or proportions using sample data and interpret the results in a business context.
- Apply bootstrapping techniques to estimate confidence intervals for non parametric statistics.

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T128_P08.py - C:/Users/Rani/Downloads/T128_P08.py (3.13.7)
File Edit Format Run Options Window Help

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Student Details
print("Name: Rani Chavre")
print("Roll No.: T128")
print("AIM: Confidence Interval Estimation")

# Load Dataset
df = pd.read_csv("Iris.csv")

# Select numerical data
data = df.select_dtypes(include=np.number).iloc[:, 0].dropna()

# Random Sample
np.random.seed(42)
sample = data.sample(30)

# 95% Confidence Interval
mean = sample.mean()
se = sample.std() / np.sqrt(len(sample))

lower = mean - 1.96 * se
upper = mean + 1.96 * se

Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on
Enter "help" below or click "Help" above for more information.

>>>

===== RESTART: C:/Users/Rani/Downloads/T128_P08.py =====
=
Name: Rani Chavre
Roll No.: T128
AIM: Confidence Interval Estimation

--- Confidence Interval ---
Sample Mean: 5.98
95% CI Lower: 5.68
95% CI Upper: 6.28

--- Bootstrap Confidence Interval ---
Bootstrap Mean: 5.98
Bootstrap CI Lower: 5.69
Bootstrap CI Upper: 6.27
```

